

Satellite Journalism:

¿why, what,
how, where?



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Satellite Journalism

1. Satellite Journalism: **why?**
2. **What** kind of stories?
3. Technicalities: **how?**
4. **Where?** Satellite data & tools
5. **Let's practice!**

Satellite Journalism: why?

- Availability → DATA SOURCE
 - Exploration → CONTEXT
 - Evidence → CORROBORATION
 - Historical series → CHANGE
 - Storytelling tool → NARRATIVE
-
- **But... satellite also have limitations!**



What kind of stories?

(adapted from *de Hoog 2021*)

1. Before/After



[This fabric is hailed as 'eco-friendly.' The rainforest tells a different story](#)

2. Geolocation/Reconstruction



The locations of Israeli soldiers and armed Palestinian men, based on video and photos collected from bystanders, journalists, security cameras and social media. Satellite image: Israeli Ministry of Housing and Construction; Graphics: The New York Times.

[The Killing of Shireen Abu Akleh: Tracing a Bullet to an Israeli Convoy](#)

What kind of stories?

(adapted from *de Hoog 2021*)

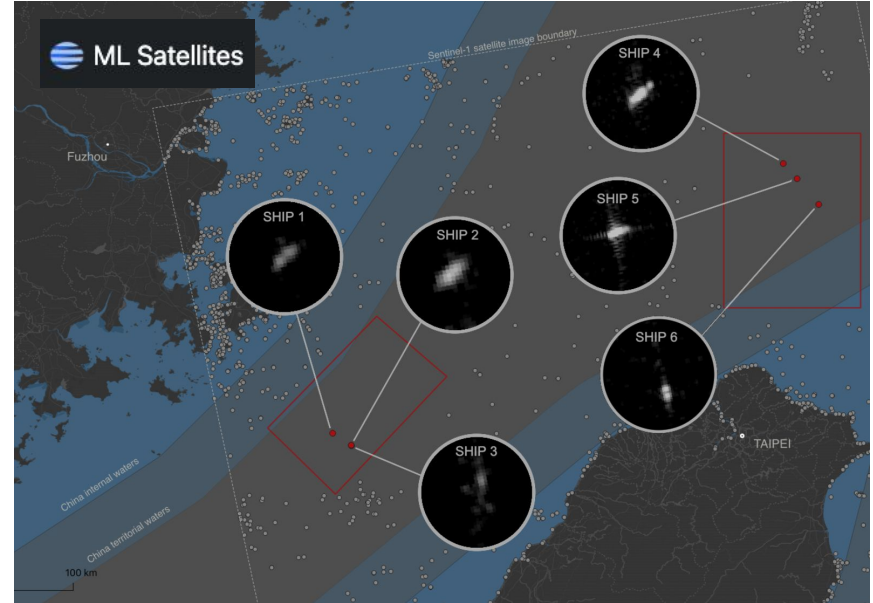
3. Smoking gun



FILE – In this July 14, 2015 satellite image provided by DigitalGlobe, two fishing trawlers load slave-caught fish onto the Silver Sea 2, center, a refrigerated cargo ship belonging to the Thai-owned Silver Sea Fishery Co., off the coast of Papua New Guinea. The Indonesian fisheries minister said the massive refrigerated cargo ship, Silver Sea 2, believed to be loaded with slave-caught fish has been seized by the navy and brought to shore after an Associated Press report alerted authorities about its presence in the country's waters. (DigitalGlobe via AP, File)

[Seafood from slaves](#)

4. Presence/absence



[When the Strait came to a halt](#)

What kind of stories?

(adapted from *de Hoog 2021*)

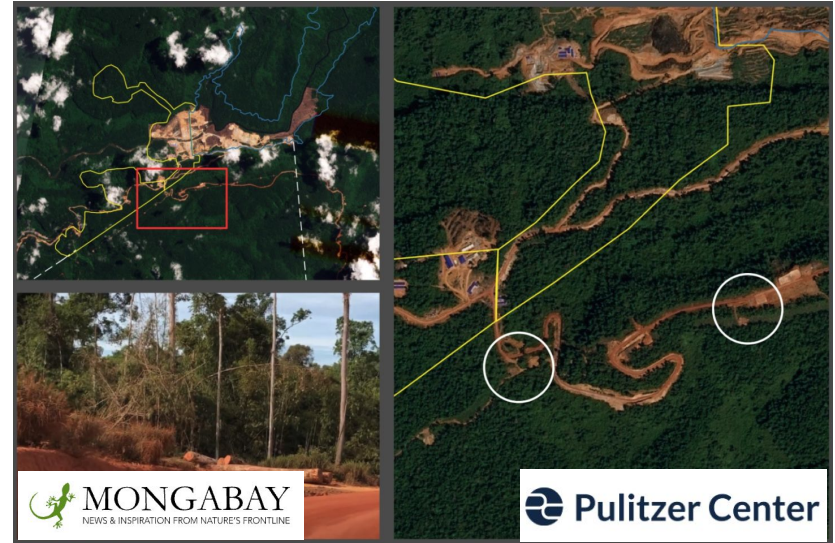
5. Context/scale



A satellite image shows the grounded cargo ship on Sunday. (Maxar/AP)

[Inside the 144-hour scramble to free the giant ship stuck in the Suez Canal](#)

6. Environmental impact

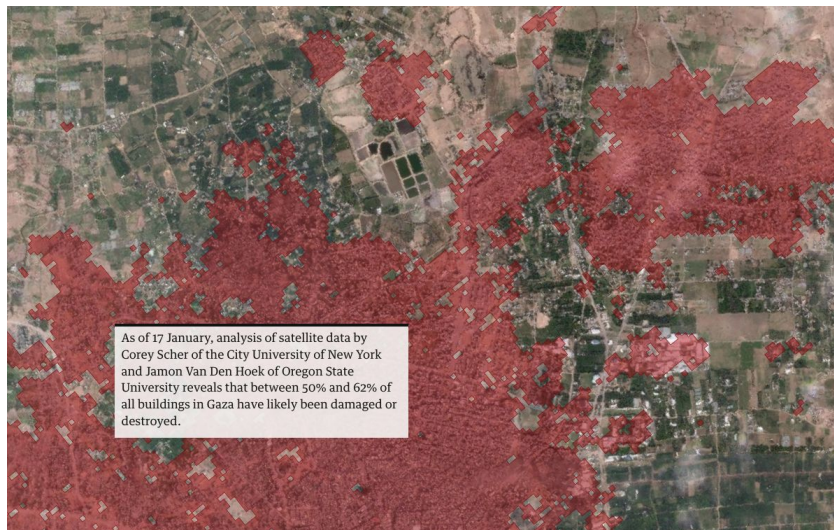


[Forest Behind Bars: Logging Network Operating Out of Cambodian Prison in the Cardamoms](#)

What kind of stories?

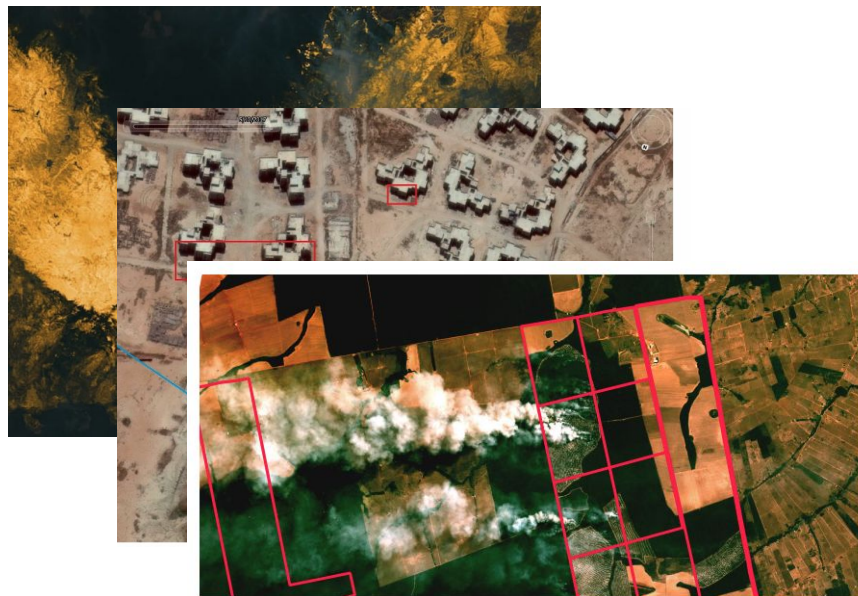
(adapted from *de Hoog 2021*)

7. Infrastructure changes



[How war destroyed Gaza's neighbourhoods – visual investigation](#)

8. Much more...!



Fires are seen burning in stretches of forest in Marçelândia, Mato Grosso, linked to soya, corn and cattle farmer Alexandra Aparecida Perinoto. © 2021 Planet Labs Inc.

Satellite Images + AI:



Amazon Mining Watch

[LINK](#)



 Pulitzer Center

AMAZON
MINING
WATCH

Amazon Mining Watch

[LINK](#)

Technicalities: how?

1. Resolution and availability
2. Common data formats
3. Satellite data is complex!
4. Tips to remember
5. **THE MUSTS**



1. Resolution and availability

Open Source/Free

Low resolution:

> 60m/pixel

Medium resolution:

10m – 30m/pixel



Aqua (MODIS)
250m Resolution



Landsat-8
30m Resolution



Sentinel-2
10m Resolution

Private providers

High resolution:

0.3m – 5m/pixel



PlanetScope (Dove)
3m Resolution



Pleiades
0.5m Resolution



Worldview-4
0.3m Resolution

(Source: Corcoran, 2018)



Venice by Sentinel 2
10 meters/pixel

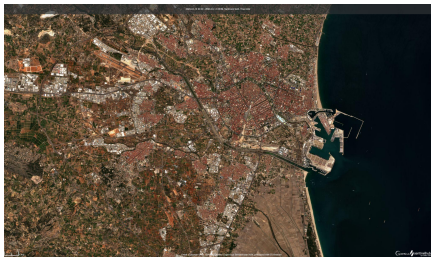
An aerial photograph of Venice, Italy, showing the dense urban layout and the Grand Canal. The image is a high-resolution satellite or aerial view, with a semi-transparent white box overlaid on the left side containing text. The text reads "Venice by AirBus" in a bold, black font, followed by "0.5 meters/pixel" in a regular black font. The background shows the intricate network of streets and buildings of Venice, with the Grand Canal winding through the center. The water of the canal is a deep blue, and the surrounding urban area is a mix of brown and grey tones, representing the dense packing of buildings and the canal's path.

Venice by AirBus
0.5 meters/pixel

2. Common data formats

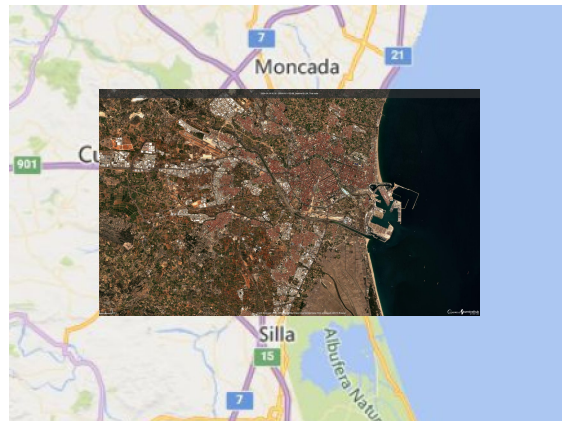
Regular Image

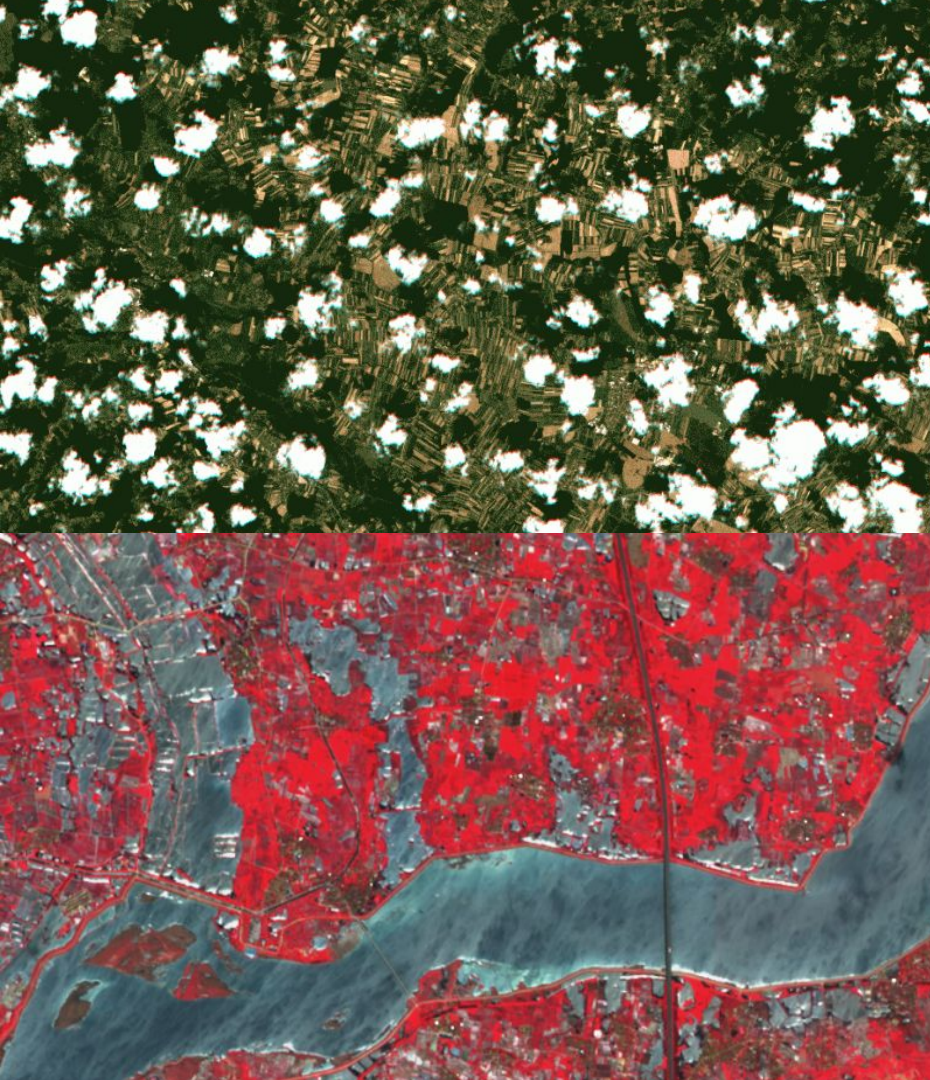
- JPG / PNG formats
- Plan image → **No geo data**
- For illustrative purposes



GeoLocated Image

- TIFF / GeoTIFF format
- Raster → **Can be geolocated**
- For GIS software (i.e. [QGIS](#)) and geospatial analysis





3. Satellite data is complex!

- Requires time & patience
- Not always is available:
 - Coverage area
 - Revisit time
 - Clouds affect images
 - [SAR as an alternative](#)
- Many bands (multi-layer):
 - RGB
 - NIR (Near-Infrared)
 - SWIR (Short-Wave)

4. Tips to remember

- Always start simple
- Don't know? Ask an expert
- Satellite are just tools
- Be skeptical
- **Experiment and explore!**





5. THE MUSTS

- Ground truth & more sources
- Tell a human story not a satellite story
 - Who is affected?
 - Voice to the voiceless
- Ethical considerations when looking from the sky
 - Accuracy
 - Privacy
 - Share your methodology

Where? Satellite data & resources

Satellite images for free

Google Earth Pro (desktop version)

<https://www.google.com/earth/versions/#earth-pro>

Sentinel Hub EO Browser

<https://apps.sentinel-hub.com/eo-browser/>

Planet Explorer

<https://www.planet.com/explore>

Google Earth Engine (to code!)

<https://earthengine.google.com/>

Satellite (provider)	Resolution	Access
Landsat (NASA)	30 m - medium	EO Browser
Sentinel-2 (ESA)	10 m - medium	EO Browser
Planet satellites	2~3m - high	Planet - NICFI Level 1

To learn more

Satellite journalism - the big picture

<https://reutersinstitute.politics.ox.ac.uk/our-research/satellite-journalism-big-picture>

Resources for Finding and Using Satellite Images

<https://giin.org/resource/resources-for-finding-and-using-satellite-images/>

How to Use Satellite Images for Investigation

<https://www.youtube.com/watch?v=5tXv0gHJMeE>

Satellite Journalism

<https://satellitejournalism.com/>

Cloud-Based Remote Sensing with Google Earth Engine

<https://www.eefabook.org/>

Artificial Intelligence, Satellites and Journalism

<https://facostarainis.github.io/ml-satellites/index.html>

Now, let's practice!

https://docs.google.com/document/d/144zZjn8FU75WSIOW8ByNn13PnDVOXqq_3ObDnlQmYes/edit?usp=sharing

Thank you!

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